

FI360 TYRE MODULE

API & INTEGRATION CONTRACT

Fleet Intelligence 360 (FI360) — Version 1.0

Fleet Associates Africa Ltd (FAA)

10 August 2026

Document Control

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Version	1.0
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Parent	FI360 Tyre Database & Data Dictionary v1.0
Core Foundation	FI360 Core Architecture & Common Services Specification v1.0
Standard	FI360 Module Development Standard v1.0
Module	FI360 Tyre Management
Owner	Fleet Associates Africa Ltd
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1. Purpose

This document defines the REST API contract and domain event schemas that expose the FI360 Tyre module's data (Document 05: FI360 Tyre Database & Data Dictionary) through the FI360 API Gateway and FI360 Core Event Service, per FI360 Core §12–13. It is the controlling interface contract for both standalone Tyre-module clients (the FI360 Tyre web/mobile UI) and future FI360 module-to-module integration (Workshop, Fleet, Parts, Telematics, Fuel, Analytics).

2. Scope

This document covers:

- REST endpoint definitions for every API area listed in the SRS (Section 21): Tyre Master, Fitment, Inspection, Movement, Lifecycle, Cost, Analytics and Integration.
- Request/response conventions: authentication, versioning, pagination, idempotency and the standard error envelope.
- The domain event catalogue (Section 22 of the SRS) with payload schema, aligned to the FI360 Core event envelope (Core §13).
- The tyre lifecycle state machine expressed as the API/event transitions that drive it.
- Cross-module integration contracts and a worked example (tyre fitment) showing the full request → event → subscriber flow.

It does not define UI/UX behaviour (Document 07) or the developer backlog and test plan (Document 08).

3. API Design Principles

- All Tyre-module APIs are accessed only through the FI360 API Gateway — never directly against the Tyre database (FI360 Core §12, Module Standard §4).
- Every protected request is authenticated (OIDC/OAuth2 bearer token) and authorized against tenant, role and action scope before it reaches the Tyre service.
- APIs are resource-oriented and map directly to the entities in Document 05; request/response payloads use the same field names as the data dictionary.
- Material state changes are transactional in the database and are followed by an asynchronous domain event — the API response does not wait for event fan-out to subscribing modules.
- APIs are versioned explicitly (/v1/...); breaking changes require a new version, per FI360 Core §18 (Service-to-Service Communication) and the Module Standard's integration rules.
- A customer using only FI360 Tyre never needs another module's API to function; integration endpoints are additive, not required for standalone operation.

3.1 Request Flow

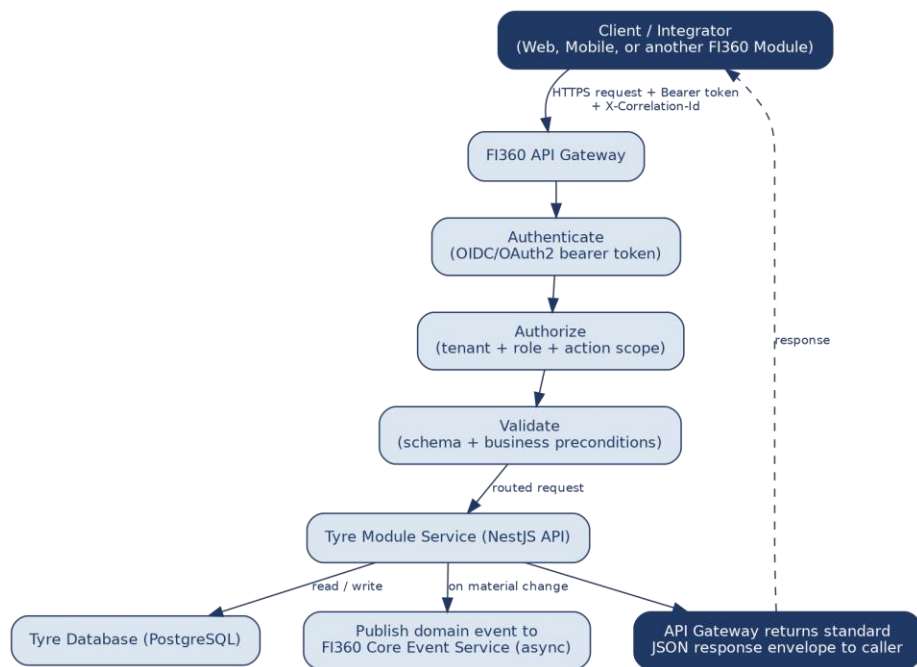


Figure 1. FI360 Tyre API request flow through the FI360 API Gateway.

4. Base Conventions

Convention	Rule
Base path	<code>https://api.fi360.example/v1/tyre/...</code>
Authentication	Authorization: Bearer <OIDC/OAuth2 token>, issued by FI360 Core Identity.
Tenant context	Resolved from the authenticated token; never accepted as a client-supplied field.
Correlation	X-Correlation-Id header on every request; propagated onto any event(s) published as a result.
Content type	application/json; charset=utf-8 for all requests and responses.

Resource (API)	Backing Table	Notes
/v1/costs	tyre_cost	Read-only consolidated ledger; populated internally from repair/retread/disposition/purchase.
/v1/tyres/{tyreId}/dispose	tyre_disposition	Terminal disposition action.
/v1/analytics/*	tyre_cost, tyre, tyre_fitment (aggregated)	Approved KPI/query endpoints (cost-per-km, availability, lifecycle).

6. API Endpoints by Area

Each table below follows the SRS's API Area grouping (Section 21). Every endpoint sits behind the request flow in Figure 1.

6.1 Tyre Master API

Method	Path	Purpose	Event(s) Published
POST	/v1/tyres	Register a new tyre.	TYRE_REGISTERED
GET	/v1/tyres/{tyreId}	Retrieve a tyre's full current record.	—
GET	/v1/tyres	Search/list tyres (filter by status, site, specification, brand).	—
PATCH	/v1/tyres/{tyreId}	Update controlled master fields (not status — see Lifecycle API).	—
POST	/v1/tyres/{tyreId}/dispose	Record final disposition (scrap/sale/trade-in).	TYRE_SCRAPPED / TYRE_RETIRED

6.2 Fitment API

Method	Path	Purpose	Event(s) Published
POST	/v1/fitments	Fit a tyre to an asset/position.	TYRE_FITTED
POST	/v1/fitments/{fitmentId}/remove	Remove a fitted tyre; closes the active fitment.	TYRE_REMOVED
POST	/v1/fitments/rotate	Rotate tyres between positions (paired remove + fit).	TYRE_ROTATED
GET	/v1/fitments	List fitment history, filterable by tyre_id or asset_id.	—

6.3 Inspection API

Method	Path	Purpose	Event(s) Published
POST	/v1/inspections	Post an inspection (tread, pressure, condition, findings).	TYRE_INSPECTED
GET	/v1/inspections	List inspection history for a tyre.	—

6.4 Movement API

Method	Path	Purpose	Event(s) Published
POST	/v1/movements	Record a tyre movement (store transfer, issue, receipt).	TYRE_RECEIVED (on initial receipt)
GET	/v1/movements	List movement history for a tyre.	—

6.5 Lifecycle API

Method	Path	Purpose	Event(s) Published
POST	/v1/repairs	Post a repair event.	TYRE_REPAIRED
POST	/v1/retreads	Dispatch a tyre for retreading.	—
POST	/v1/retreads/{retreadId}/receive	Record return of a retreaded casing; increments retread_count.	TYRE_RETREADED
GET	/v1/repairs, /v1/retreads	List repair/retread history.	—

6.6 Cost API

Method	Path	Purpose	Event(s) Published
GET	/v1/costs	List consolidated cost-ledger entries for a tyre.	—
GET	/v1/costs/summary	Lifecycle cost and cost-per-kilometre summary for a tyre or fleet.	—

6.7 Analytics API

Method	Path	Purpose	Event(s) Published
GET	/v1/analytics/availability	Fleet-wide tyre availability/utilisation KPI.	—
GET	/v1/analytics/cost-per-km	Cost-per-kilometre by tyre, brand, site or asset.	—
GET	/v1/analytics/lifecycle	Average lifecycle duration/distance by specification.	—

6.8 Integration API

Method	Path	Purpose	Event(s) Published
GET	/v1/integration/tyres/by-asset/{assetId}	Resolve current tyre positions for a given FI360 Asset (used by Fleet/Workshop).	—

Method	Path	Purpose	Event(s) Published
GET	/v1/integration/events/{eventId}	Retrieve a previously published event payload by ID (replay support).	—

7. Tyre Lifecycle: API-Driven State Machine

Every status transition on tyre.status_id (Document 05, Section 9) is driven by exactly one of the endpoints above, and every transition publishes a corresponding domain event. This keeps the lifecycle auditable end to end: API call → database transaction → event.

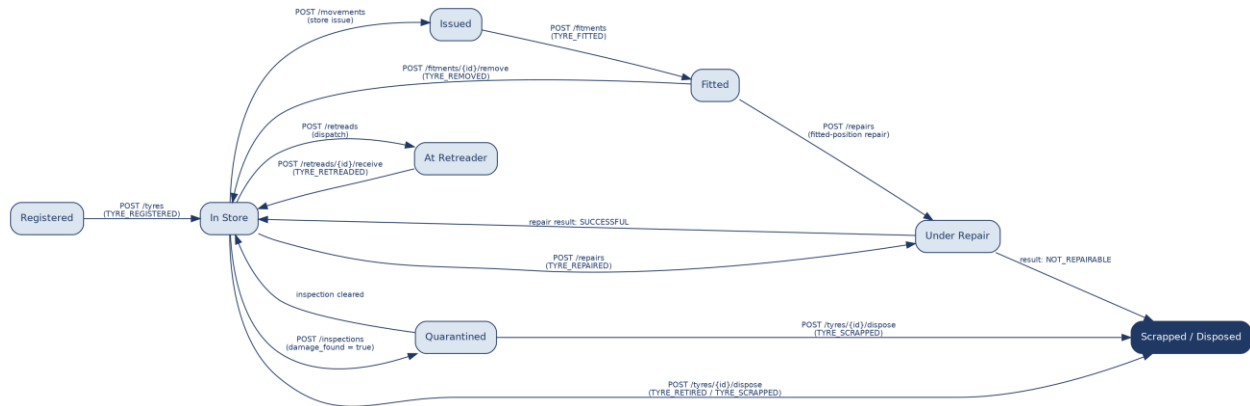


Figure 2. Tyre lifecycle state machine with the API call and event that drives each transition.

8. Domain Events

Events follow the FI360 Core event envelope (Core §13): eventId, eventType, version, tenantId, entityId, timestamp, source, correlationId, payload. The Tyre-specific eventType values and their payloads are:

Event Type	Published When	Key Payload Fields
TYRE_REGISTERED	A new tyre is successfully registered.	tyreId, specificationId, tyreTypeId, statusId
TYRE_RECEIVED	A tyre receipt/goods-in movement is posted.	tyreId, siteId, supplierId, receivedDate
TYRE_FITTED	A fitment is posted.	tyreId, assetId, positionCode, fittedDate, fittedOdometer
TYRE_REMOVED	A removal is posted, closing the active fitment.	tyreId, assetId, positionCode, removedDate, removalReasonId
TYRE_ROTATED	A rotation (paired remove+fit) completes.	tyreId, fromPositionCode, toPositionCode, assetId, rotationDate
TYRE_INSPECTED	An inspection is posted.	tyreId, inspectionDate, treadDepthMm, pressureKpa, conditionCodeId, damageFound
TYRE_REPAIRED	A repair record is posted.	tyreId, repairTypeId, resultCode, costAmount, currencyCode
TYRE_RETREADED	A retread cycle is received back and completed.	tyreId, retreadCycleNumber, resultCode, newTreadDepthMm

Event Type	Published When	Key Payload Fields
TYRE_RETIRED	A tyre is retired from active use (non-scrap disposition).	tyreId, dispositionDate, reasonId
TYRE_SCRAPPED	A tyre is scrapped.	tyreId, dispositionDate, reasonId, disposalValue

8.1 Event Publish and Subscribe Flow

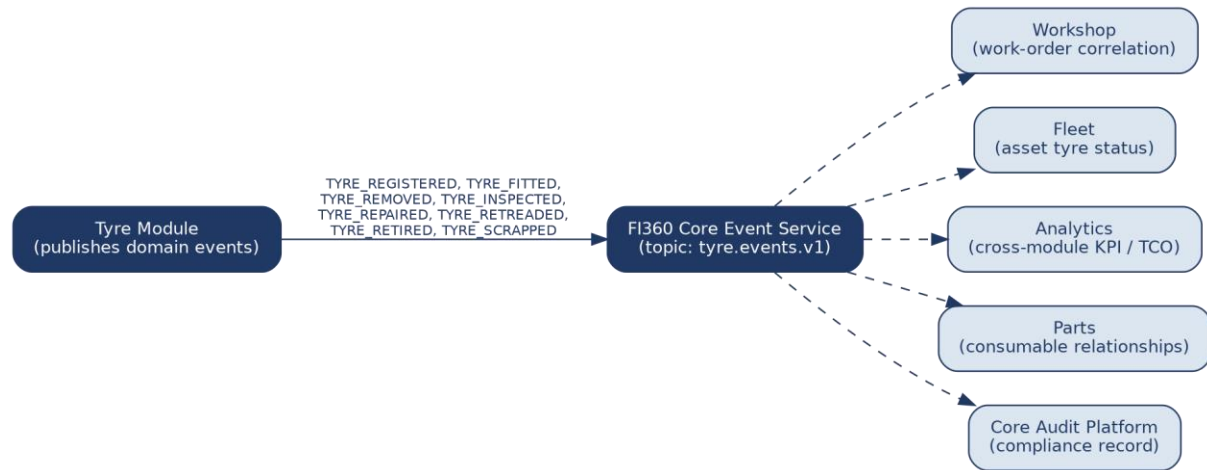


Figure 3. Tyre domain events are published once to the FI360 Core Event Service, which fans them out to every subscribing module — the Tyre module has no knowledge of who is listening.

9. Cross-Module Integration Contracts

Per the SRS (Section 23) and FI360 Core §17, the Tyre module never queries another module's database directly. All cross-module context is exchanged through the canonical Asset reference, the API endpoints above, and the events in Section 8.

Module	Integration Purpose	Mechanism
Fleet	Asset ID, vehicle context, mileage/hours.	Tyre reads/writes reference core.asset (FI360-AST-#####) only; consumes Fleet-published mileage/hours events where available.
Workshop	Work orders and tyre-related maintenance activities.	Workshop subscribes to TYRE_FITTED / TYRE_REMOVED / TYRE_REPAIRED; may call GET /v1/integration/tyres/by-asset/{assetId}.
Parts	Tyre/related stock or consumable relationships.	Parts subscribes to tyre events where consumables (valves, patches) are tracked; no direct schema coupling.
Telematics	Mileage/hours/vehicle usage signals.	Telematics-published usage events are consumed by the Tyre module to enrich tyre_fitment odometer/hour fields where configured.
Fuel	Potential correlation with vehicle operating cost.	Analytics-layer correlation only; no direct Tyre↔Fuel API dependency.
Analytics	Cross-module KPI and TCO analysis.	Analytics consumes all Tyre events plus GET /v1/analytics/* endpoints.

9.1 Worked Example: Tyre Fitment Across Modules

The following trace shows a single fitment transaction from UI action through to every module that may react to it. Steps 8a–8c only occur for tenants that have also purchased the named module; a Tyre-only tenant simply stops after step 7.

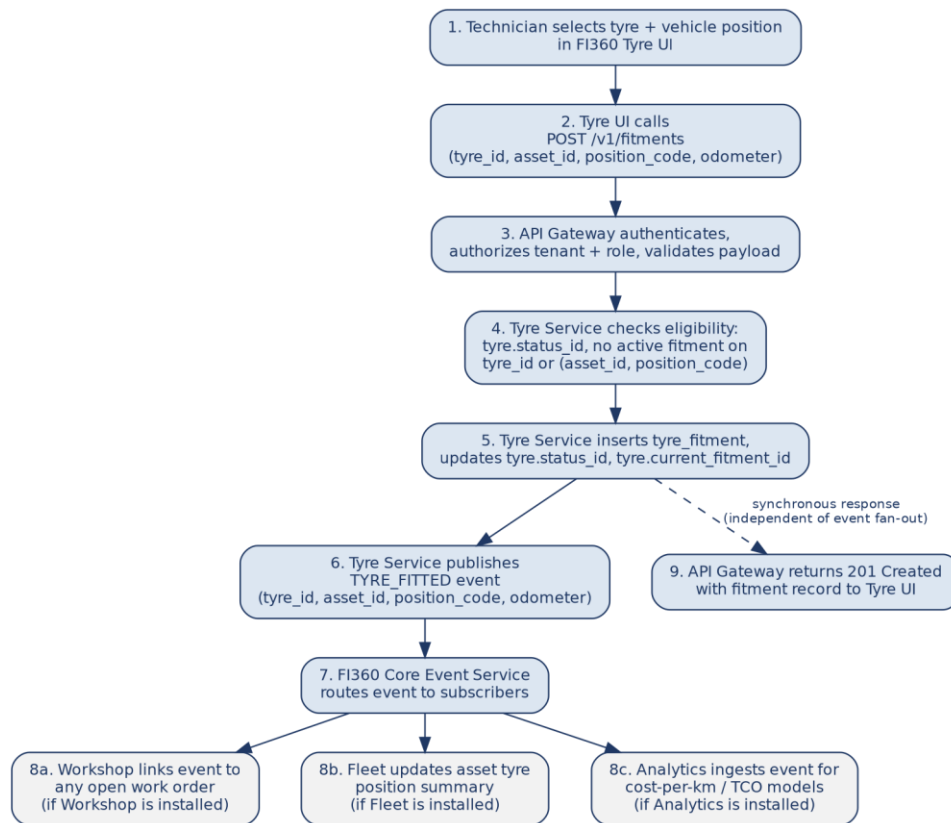


Figure 4. End-to-end trace of a tyre fitment: synchronous API path (steps 1–5, 9) and asynchronous event fan-out (steps 6–8).

10. Authentication & Authorization

- Authentication is delegated entirely to FI360 Core Identity (OIDC/OAuth2); the Tyre module never issues or validates credentials itself.
- Authorization is role- and scope-based: e.g. tyre:read, tyre:fitment:write, tyre:dispose, each assignable per role at tenant/site/department scope per FI360 Core §6.2.
- Deny-by-default applies to all protected endpoints; a missing or insufficient scope returns 403 with error.code = FORBIDDEN.
- Service-to-service calls (e.g. Analytics pulling /v1/analytics/*) authenticate using a service identity issued by FI360 Core, not a user token.

11. Versioning & Compatibility

- All endpoints are namespaced under /v1/; breaking changes are introduced under /v2/ with both versions supported through a defined deprecation window.
- Event schemas carry an explicit version field; consumers must tolerate additive (non-breaking) payload changes within the same version.
- Reference/lookup codes (Section 8.12 of Document 05) are stable identifiers — a code's label may change, but its meaning on historical records does not, per FI360 Core §8.

12. Rate Limiting, Idempotency & Error Handling

Concern	Rule
Rate limiting	Enforced at the API Gateway per tenant and per client; standard 429 Too Many Requests with Retry-After header on breach.
Idempotency	POST endpoints that create a resource require Idempotency-Key; a repeated key with an identical payload returns the original 201 response rather than creating a duplicate.
Validation errors	422 Unprocessable Entity with error.details listing each failing field.
Business-rule rejections	409 Conflict for state conflicts (e.g. fitting an already-fitted tyre), with a specific error.code such as TYRE_ALREADY_FITTED.
Not found	404 Not Found for unknown business IDs; the internal UUID is never exposed in the error body.
Server errors	500 Internal Server Error with only error.correlationId returned to the client; details are logged server-side, not exposed.

13. Acceptance Criteria

Area	Acceptance Condition
Contract completeness	Every table in Document 05 is reachable through at least one documented endpoint.
Traceability	Every state-changing endpoint publishes the event named against it in Section 6, carrying the correlationId from the originating request.
Isolation	No endpoint accepts a client-supplied tenant_id; tenant context is always derived from the authenticated token.
Idempotency	Replaying a POST with the same Idempotency-Key never creates a duplicate row.
Standalone operation	All endpoints in Sections 6.1–6.6 function correctly for a tenant with no other FI360 module installed.
Event fan-out	A tenant with Workshop, Fleet and/or Analytics installed receives every relevant event without any change to Tyre-module code.

14. Next Document

DOCUMENT 07 — FI360 TYRE UI/UX SPECIFICATION will define the screens, navigation and interaction patterns that consume the endpoints in this contract, including the registration, fitment, inspection and dashboard workflows referenced throughout this document.

FI360 Tyre principle: every tyre has one identity, one traceable lifecycle and one auditable history.